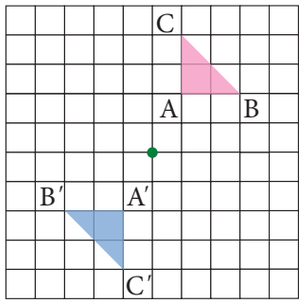
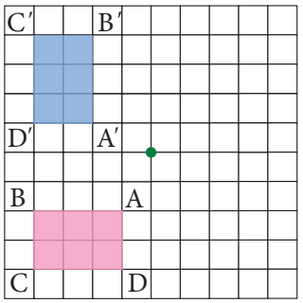
**Solution:**

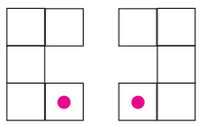
(i) 180°

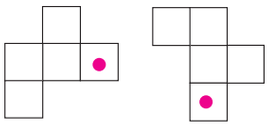


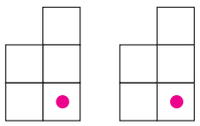

(ii) 90° counter clockwise




**Example 4.7** Describe the transformation involved in the following pair of figures. Write translation, reflection or rotation.

(i) 

(ii) 

(iii) 

**Solution:**

(i) Reflection (ii) Rotation (iii) Translation

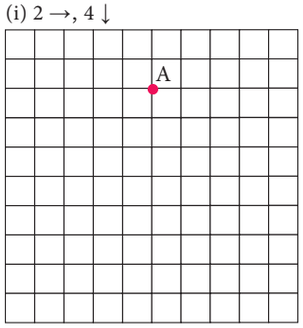
DO YOU KNOW?

A *glide reflection* is a combination of two transformations: a reflection about a line and a translation. Here the translation is parallel to the line of reflection. Reversing the order of the composition will not affect the outcome. We can translate first and then reflect, or reflect first and then translate.

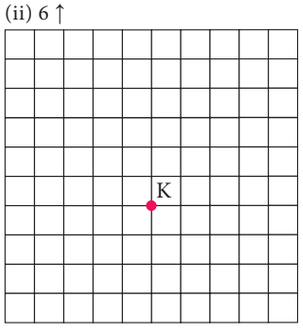
## Exercise 4.1

1. Find the new position of each point using the translation given.

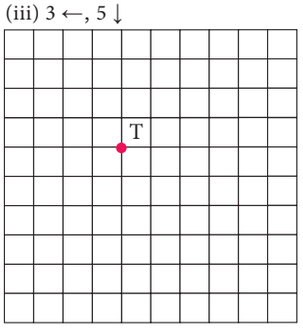
(i) 2 →, 4 ↓



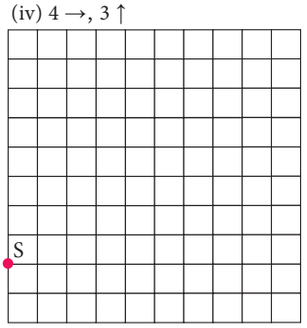

(ii) 6 ↑




(iii) 3 ←, 5 ↓

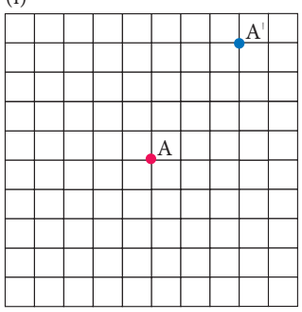



(iv) 4 →, 3 ↑

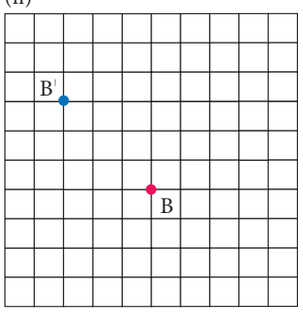



2. How is the pre-image translated to the image?

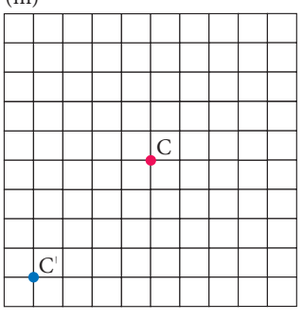
(i)



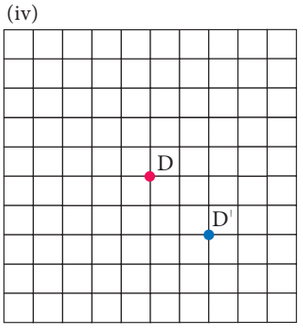

(ii)




(iii)




(iv)






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